

InsightAI Dashboard

Innovative web application for interactive data visualization and analysis with Artificial Intelligence.

Group:

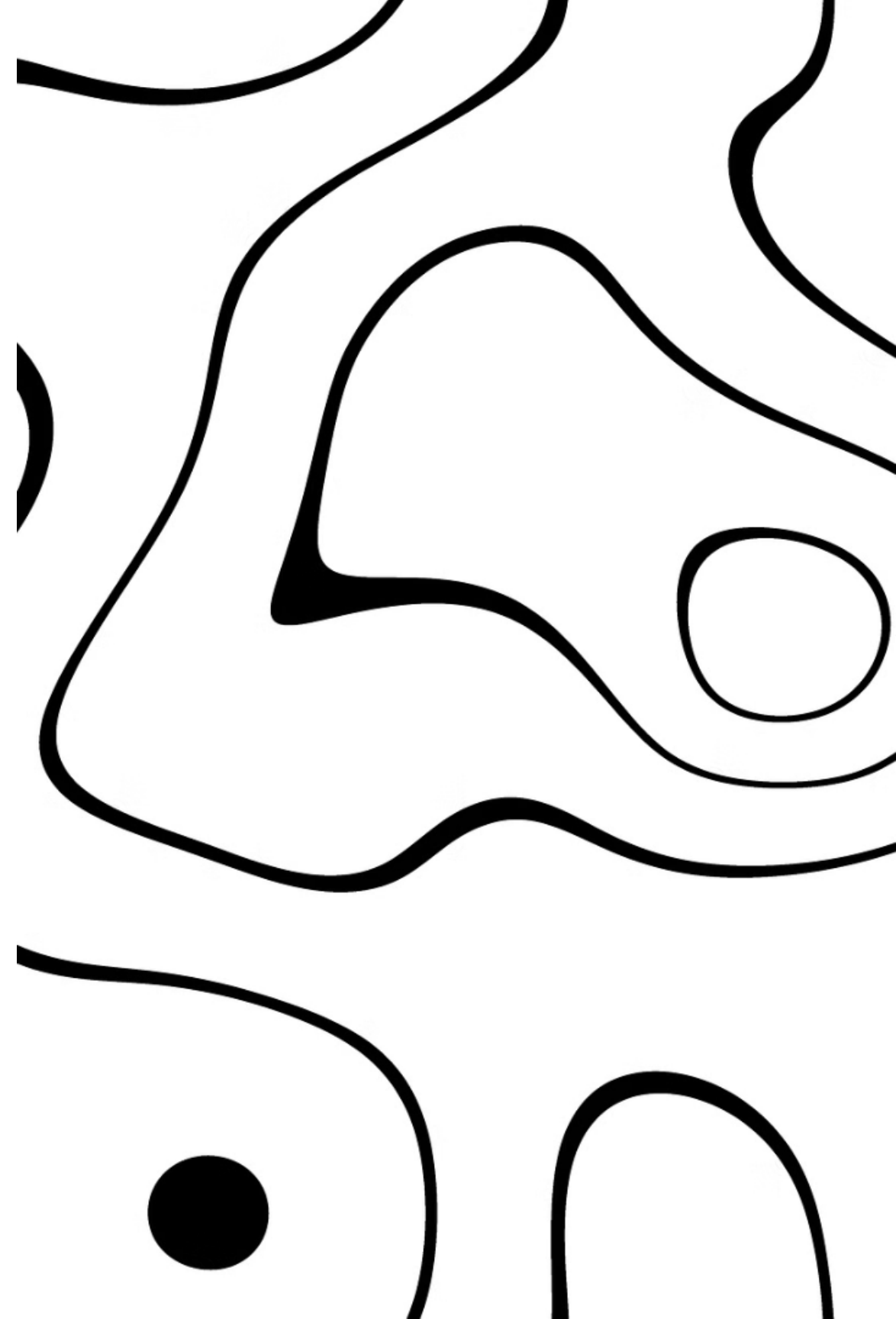
Sohaib Ahmed 21L-5635

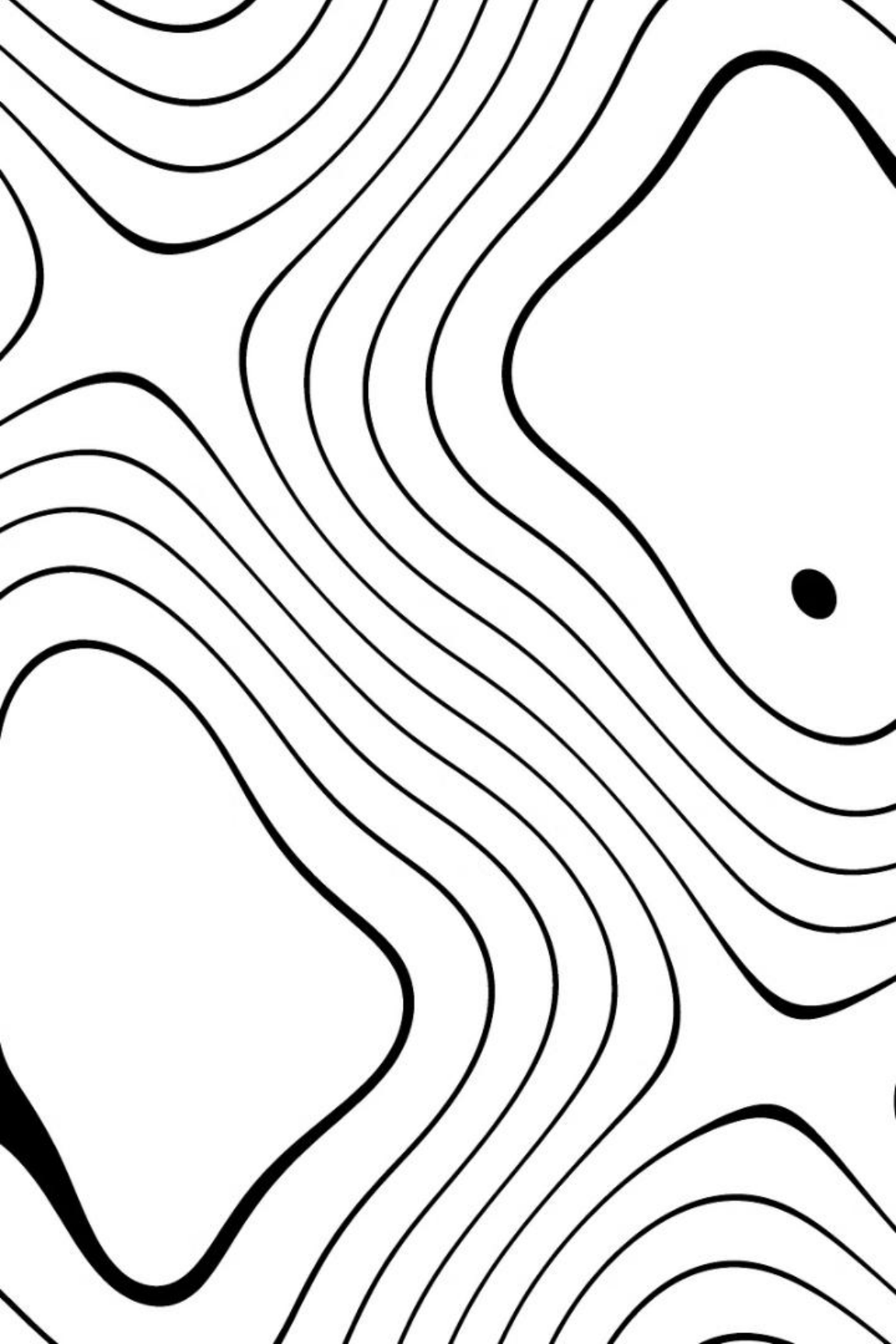
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Muhammad Zeeshan Nazar





Problem Statement

Complex Applications

Limited flexibility hinders data exploration.

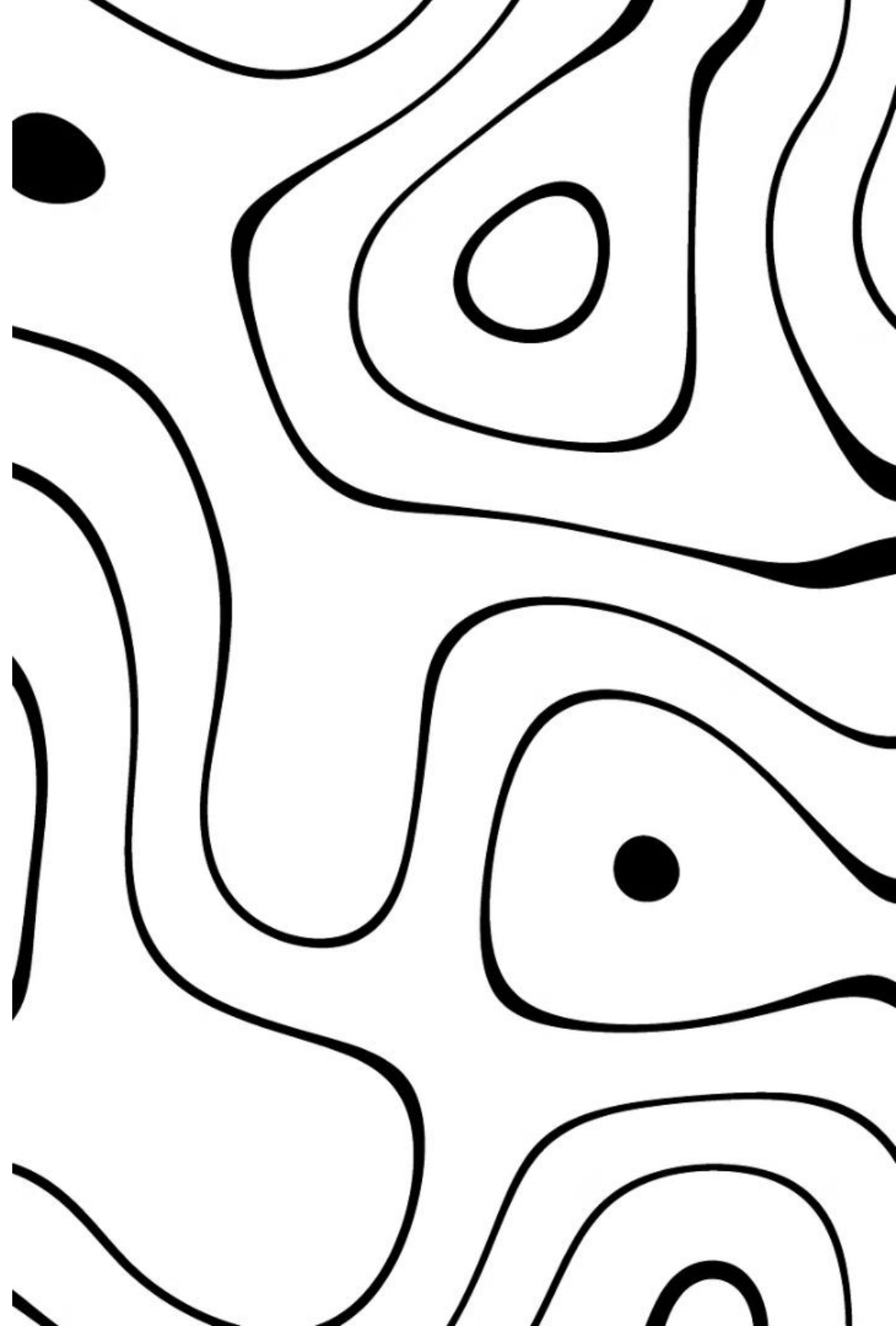
Technical Barriers

Non-technical users struggle with analysis tools.

Costly Solutions

Extracting meaningful insights is challenging.

Project Scope



In-Scope

1

Data Ingestion

Support for structured and unstructured data formats

2

Analysis Tools

AI-powered and manual data manipulation options

3

Visualization

Creation of interactive charts, graphs, and diagrams

4

Reporting

Generation of custom reports and actionable insights

5

Diagram Creation

Flowcharts, ER diagrams, and more

6

User-Friendly Interface

Responsive design for all expertise levels

7

Data Cleaning Tools

Preprocessing options for dataset optimization

8

Custom Dashboards

Create and export dashboards easily

9

AI Powered Dashboards

Create and export AI generated dashboards easily

Out-of-Scope

1

Real-Time Data Connections

No live data streaming support

2

Mobile Development

No mobile app versions planned

3

Collaborative Editing

No simultaneous multi-user editing

4

Advanced AI Features

Advanced AI features deferred for future

5

Enterprise-Level Features

Large-scale enterprise solutions not included

Goals Achieved



File Controls

Efficient handling of file uploads, storage and format compatibility.



Authentication

Secure user registration and login system to manage access and ensure data privacy.



Chart Rendering

Dynamic creation of charts and graphs for clear data representation.



Dashboard

Central hub displaying key insights and user-customized data visualizations.

Goals Achieved



Data Table Page

Organized representation of uploaded datasets for easy viewing and interaction.



Data Visualization Agent

AI-driven insights enabling users to generate advanced visual analytics.



Research Assistant Agent

Develop an assistant capable of summarizing data, generating insights and answering user queries intelligently.



SQL and Mermaid Diagram Agent

Tool for executing SQL queries to analyze datasets and automatically generating flowcharts and diagrams for structured visualization.

Related Applications

Databox

- Lacks AI-driven analytics
- Limited metrics
- No manual data handling

Geckoboard

- Complex interface
- Limited large dataset handling
- High pricing

Tableau Public

- No AI features
- Data is public
- High costs for full version

InsightAI Dashboard

- AI insights generation
- Automated data handling
- Support for unstructured files
- Handling of larger uploads
- Research assistant and relational diagrams functionality

System Overview

1

Web Application

User-friendly interface with customizable themes.

2

Data Processing

Combining Manual methods with AI-powered analysis techniques.

3

Visualization Engine

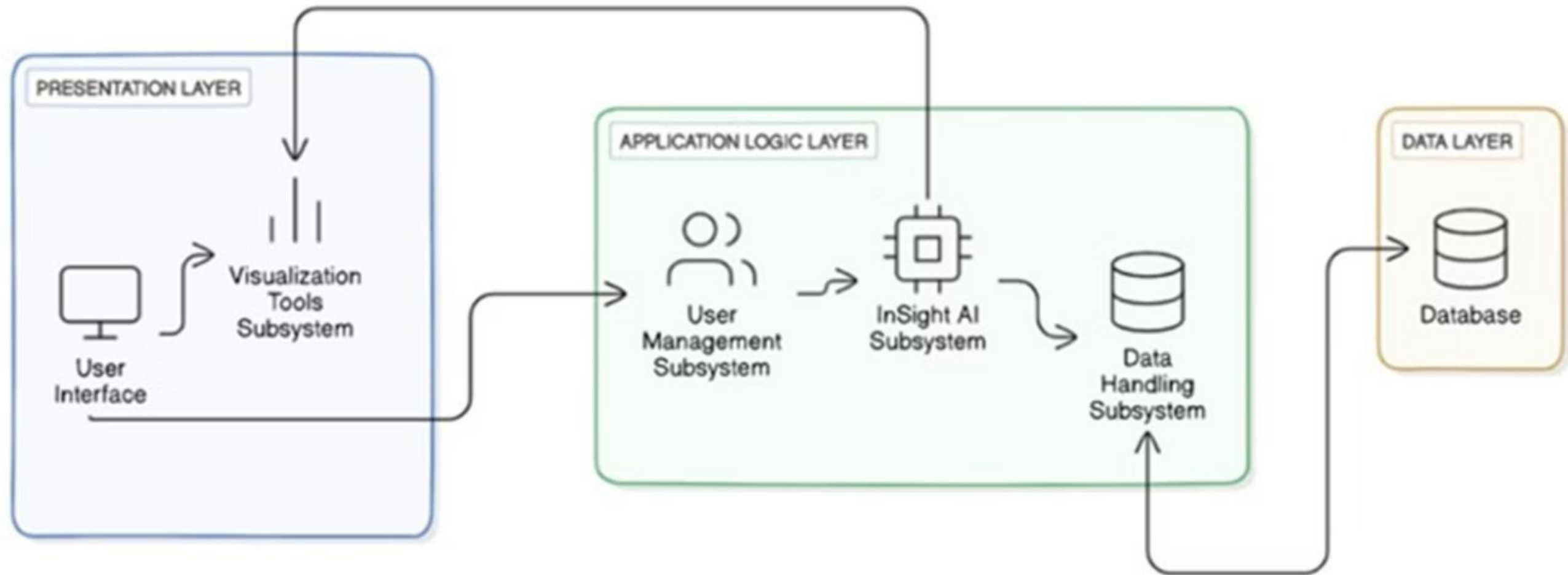
Interactive tools for creating charts and diagrams.

4

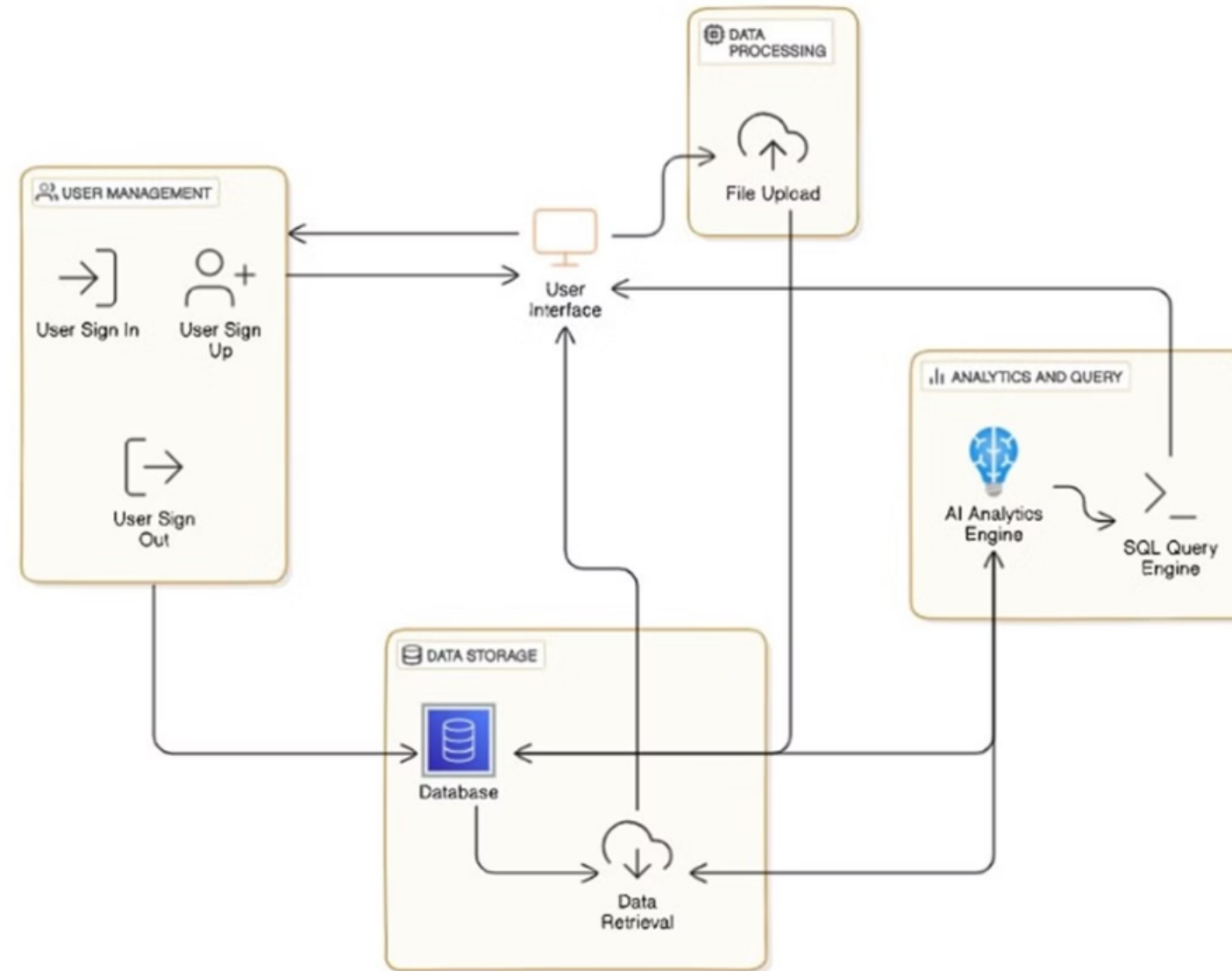
AI Assistant

Engage with AI for Advanced Data Insights

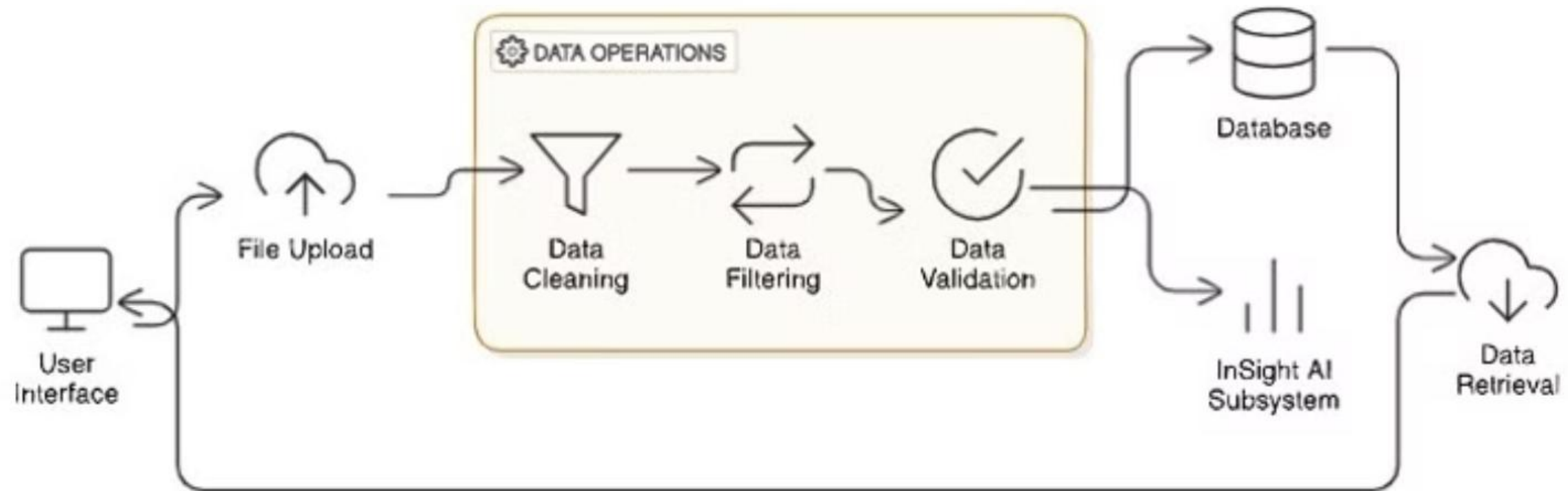
High Level System Architecture



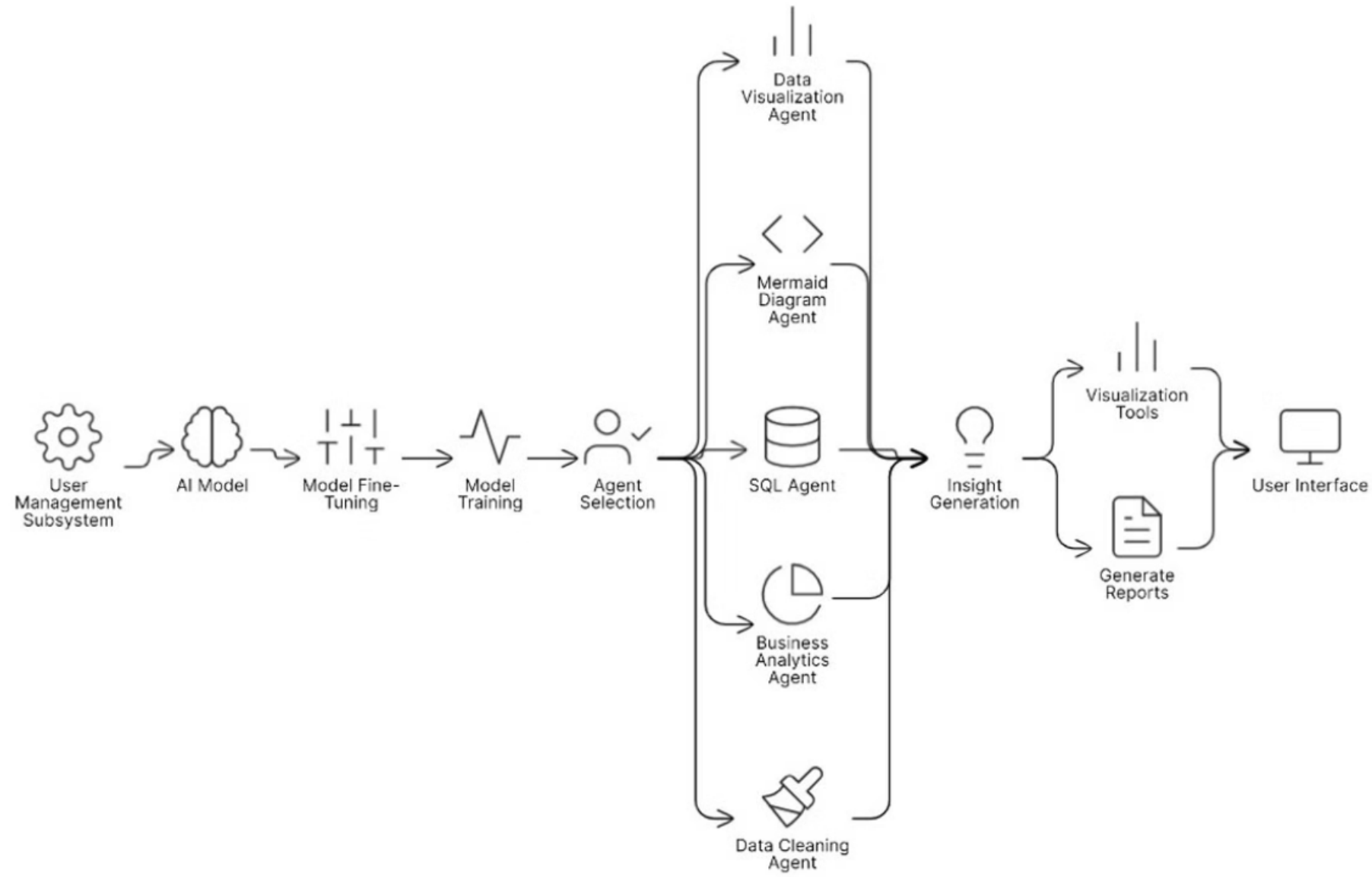
User Management Subsystem



Data Handling Subsystem



InsightAI Engine Subsystem



Important Requirements

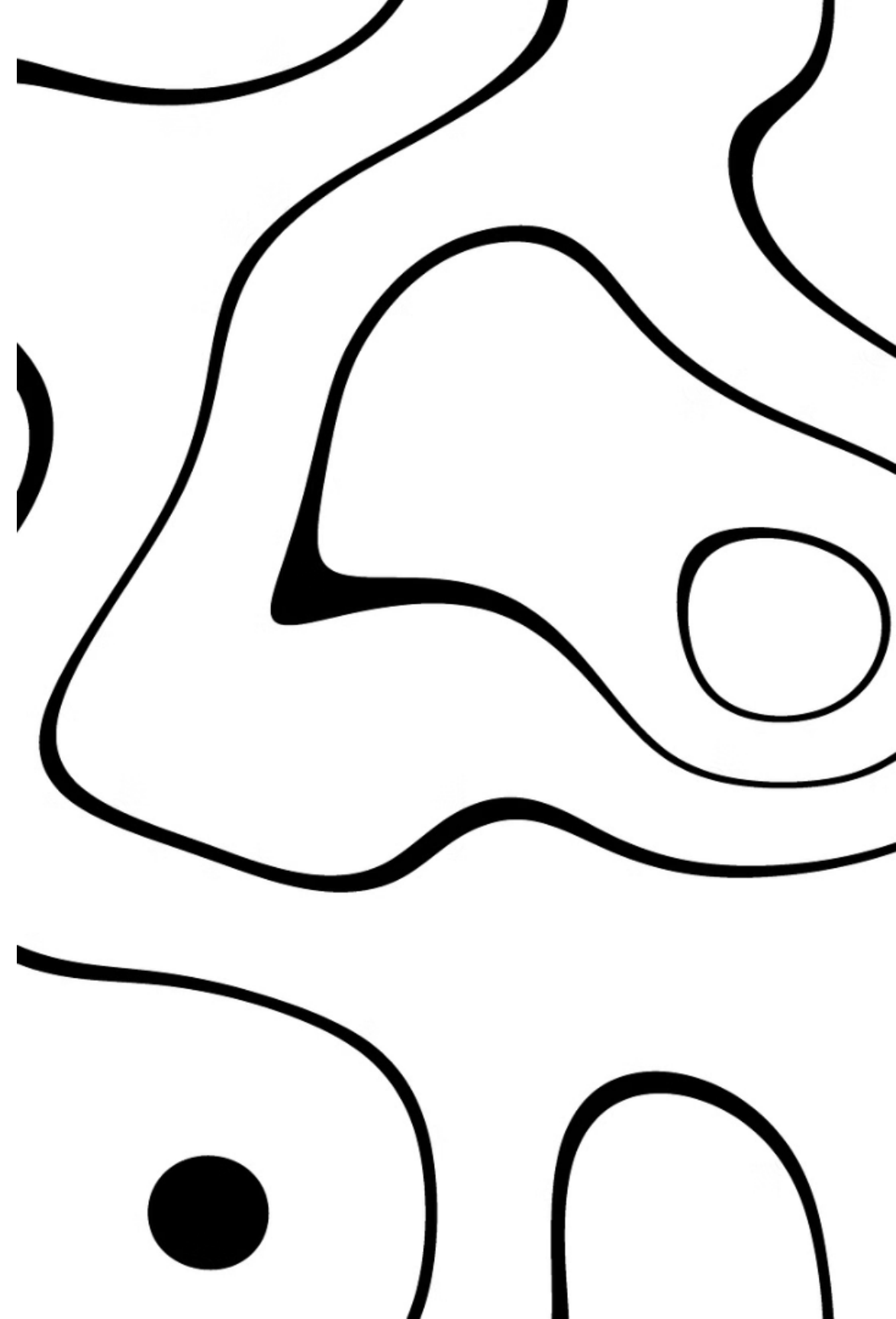
Functional requirements

1. User Registration and Login
2. Support various data formats.
3. Data Analysis with AI and manual techniques.
4. Data Visualization
5. Report Generation
6. Customized Dashboards

Non-functional requirements

1. Guarantee reliability and uptime.
2. Ensure Data Security
3. Scalability for Large Datasets and User Growth
4. User-Friendly Interface for All Users
5. Cross-Platform Compatibility
6. Performance Optimization

Graphical User Interface



Welcome

Log in or Create account to get back to your dashboard!

Create your account

Welcome! Please fill in the details to get started.



Continue with Google

or

Username

Email address

Password



Continue ▶

Already have an account? [Sign in](#)



Welcome

Log in or Create account to get back to your dashboard!

Sign in to Dashboard

Welcome back! Please sign in to continue



Continue with Google

or

Email address or username

Continue ▶

Don't have an account? [Sign up](#)



Unlock the Power of AI for Your Data

From data handling to relational insights—all automated for you.

[Get Started](#)

Why InsightAI Dashboard?



Manual Data Handling

Manual Data Handling



Upload Size (Up to 500MB)

Upload Size (Up to 500MB)



Chat with Data

Chat with Data



Automated Data Cleaning

Automated Data Cleaning



Relational Diagrams

Relational Diagrams



SQL Execution

SQL Execution



Customizable Dashboards

Customizable Dashboards



AI-Powered Insights

AI-Powered Insights

How InsightAI Works



1. Upload Your Data

Securely upload large datasets, including structured and unstructured formats.



2. AI Automates Insights

Clean, structure, and visualize data effortlessly with our AI-powered tools.



3. Create Custom Dashboards

Build intuitive, actionable dashboards for data-driven decision-making.

Tailored for Every Industry



Healthcare

Drive smarter patient data insights for improved care and operations.



Finance

Visualize trends and forecasts in real time for better financial decisions.



Marketing

Optimize campaigns with deep data insights and customer behavior analysis.



Retail

Analyze sales, trends, and customer behavior to boost your retail strategy.

Data Hub

File Selection

Upload, select, or manage your data files

Upload File

Delete Files

iris.csv

Selected: iris.csv

Clear selection

File Content

Create New

Version History

Compare Versions

Column Calculator

Rename File

Reset All

Analyze Data

☐	sepal_length	 	sepal_width	 	petal_length	 	petal_width	 	species	 
☐	5.1	 	3.5	 	1.4	 	0.2	 	setosa	 
☐	4.9	 	3	 	1.4	 	0.2	 	setosa	 
☐	4.7	 	3.2	 	1.3	 	0.2	 	setosa	 
☐	4.6	 	3.1	 	1.5	 	0.2	 	setosa	 



Select Dashboard



New Dashboard Name

+ Create

📷 Save

Select and resize nodes by dragging the handles



Welcome ,Hassan

Great to see you!

Data Hub

File Selection

Upload, select, or manage your data files

Upload your Files here!



Add new files



Drag & Drop or Choose files to upload

CSV, XLSX, DB, TXT, TSV, PDF, XML

Cancel

Import

 Upload File

 Delete Files



No files available

Upload a file to get started

Welcome Hassan!

Great to see you!

Create Powerful Data Dashboards

Design Interactive Dashboards, Analyze Your Data, and Uncover Insights

Get Started

Welcome ,Hassan

Great to see you!



Select Dashboard



New Dashboard Name

+ Create

 Save

Select and resize nodes by dragging the handles



Welcome Hassan!

Great to see you!

InsightAI Data Analysis

Select Data Source

Upload a new file or select from your existing files

Upload New File

Existing Files

Select a file



Welcome Hassan!

Great to see you!

InsightAI Data Analysis

Select Data Source

Upload a new file or select from your existing files

Upload New File

Existing Files

 iris.csv (csv)

 **iris.csv**
Type: CSV • ID: 6

Select Data Source

Upload a new file or select from your existing files

Upload New File

Existing Files

 iris.csv (csv)

 iris.csv
Type: CSV • ID: 6

Analysis Options

Configure how you want to analyze your data

☐ Generate Comprehensive Report

☐ Generate Flow Diagram 

Ask Questions About Your Data

Ask specific questions about your data

Your Question

e.g., Clean this data and suggest ML models



Suggested questions:

Show me a summary of this data

Create a bar chart of the top 5 items

What trends do you see in this data?

Calculate the average values by category

Clean this data and suggest ML models

Welcome Hassan!

Great to see you!

InsightAI Data Analysis

Select Data Source

Upload a new file or select from your existing files

Upload New File

Existing Files

Select a file



Welcome Hassan!

Great to see you!

InsightAI Data Analysis

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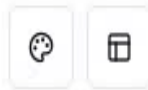
Upload New File

Existing Files

Select a file



Welcome, Hassan! Chat with Your Data



Mermaid Diagram ▾



Type your message...



Constraints and Limitations

1

Data Size

Limited processing for extremely large datasets.

2

Complex Queries

Performance issues with highly complex SQL queries.

3

AI Accuracy

Potential for errors in AI-generated responses.

4

Connectivity

Dependency on internet connection for full functionality.

Conclusion: Transforming Data Analysis

Innovation

Bridging gap between complex data and user-friendly analysis.

Empowerment

Enabling all users to gain valuable data insights.

Future-Ready

Positioned for growth with AI and visualization capabilities.